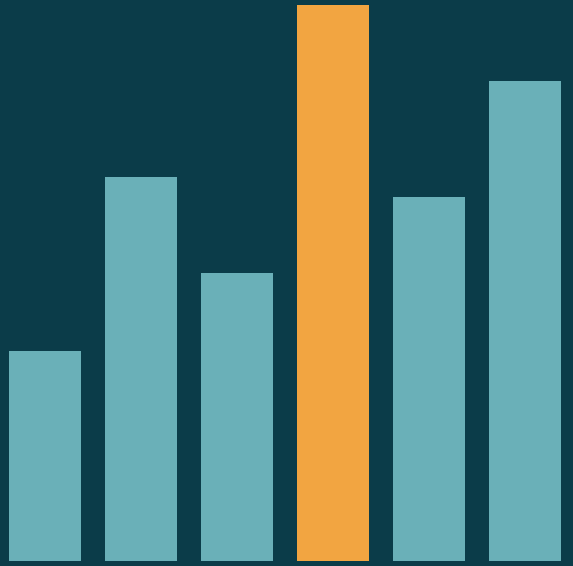
ADSAA Algorithm Visualizer

An interactive web platform for learning algorithms and data structures

Pragati Engineering College (A)

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Project Overview

The ADSAA Algorithm Visualizer is a web-based educational platform that helps students understand algorithms and data structures through step-by-step visual animations.

Project Team

Md. Zahurunnisa Begum	25A31A05EI
P. Keerthana	25A31A05EN
J. Manideep	25A31A05FL
G. Supradeep	25A31A05FD

5
Sorting algorithms

2
Searching algorithms

3
Graph algorithms

3
Data structures

Problem and Solution

The Problem

- Learning from textbooks and static diagrams
- Manually tracing code is slow and error-prone
- Hard to see comparisons, swaps and traversal steps

Our Solution

- Interactive, animated algorithm visualisation
- Custom input and adjustable animation speed
- Live statistics for comparisons and swaps

Objectives



Provide an interactive platform for learning



Visualize sorting algorithms step by step



Help students understand searching techniques



Introduce graph and shortest-path concepts



Cover basics of stacks, queues and trees



Display comparisons and swaps statistics

Learning Modules



Sorting

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort



Searching

- Linear Search
- Binary Search



Graphs

- BFS
- DFS
- Dijkstra's



Data Structures

- Stack
- Queue
- Binary Tree

Key Features

Custom Input

Enter comma-separated numbers to test any sequence.

Speed Control

Slow down or speed up the animation as you learn.

Live Statistics

See comparisons and swaps or moves as the sort runs.

Complexity Info

Time and space complexity shown for each algorithm.

Step-by-Step View

Watch every operation instead of only reading code.

Browser-Based

Runs in any modern browser on GitHub Pages, no install.

Technologies Used



HTML

Structure and content of the pages



CSS

Layout, colours and visual styling



JavaScript

Interactivity, algorithm logic and animation



Git

Version control for the team



GitHub Pages

Hosting the public website

System Architecture

Browser Environment



User Interface

Navigate modules and enter input



Algorithm Module

Runs the selected algorithm's operations



Visualization Module

Shows the execution visually



Statistics Module

Displays comparisons and swaps

Application Workflow



Open Website

Launch the hosted visualizer

Select Module

Sorting, searching, graphs or data structures

Enter Input

Provide numbers, nodes or values

Execute

Run the chosen algorithm

Visualize

Watch steps, results and statistics

Sorting visualizer: choose an algorithm, enter comma-separated numbers, set the speed, then start and watch the bars move.

Test Cases

#	Test case	Expected result
1	**Website Access**	Opens in a supported browser
2	**Module Navigation**	Learning modules can be selected
3	**Algorithm Selection**	A sorting algorithm can be chosen
4	**Numerical Input**	Formatted numbers are accepted
5	**Sorting Execution**	The sorting process is displayed
6	**Animation Speed**	Speed control changes the animation
7	**Statistics Display**	Comparisons and swaps are shown

Results are recorded after running each test on the live website.

Advantages and Limitations

Advantages

- Interactive and beginner-friendly
- Many algorithm categories on one site
- Custom input, speed control, live statistics
- Useful for self-study and classroom demos

Limitations

- Needs basic programming knowledge
- Covers only the implemented algorithms
- Needs a browser and internet access
- Does not replace proofs or deep analysis

Future Scope

1 More Algorithms

Heap Sort, Counting Sort and minimum spanning trees

2 Step Controls

Forward, backward, pause and reset

3 Code Display

Pseudocode and code in C, C++, Java and Python

4 Complexity Graphs

Compare best, average and worst cases

5 Quizzes

Practice exercises and progress tracking

6 Better Access

Mobile-friendly and accessible design

Conclusion



Learn by seeing

Algorithm steps become visual, interactive and easier to follow.



One platform

Sorting, searching, graphs and data structures together.



Built to grow

More algorithms, code views and quizzes are planned.

Thank You

Questions and discussion

Live demo

zyntrixtechteam.github.io/ADSAA-ALGORITHM-VISUALIZER

